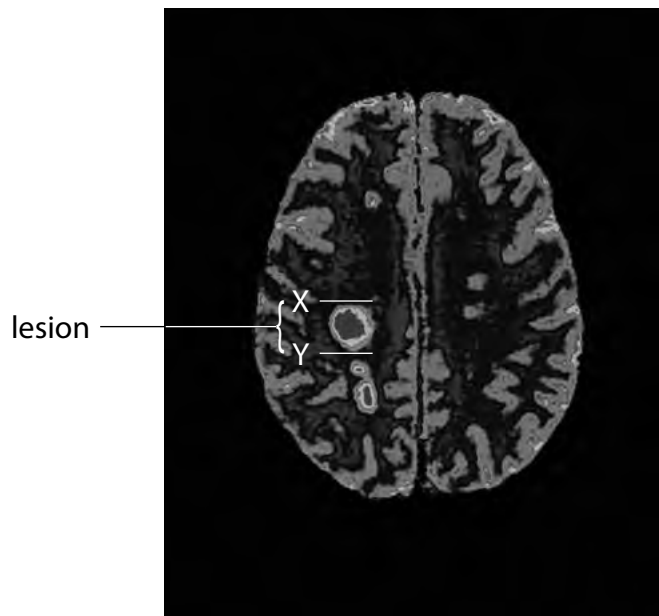


- 5 Multiple sclerosis (MS) is a disease that causes the destruction of the myelin sheaths around neurones.

This destruction is caused by the person's own immune system.

Lesions form in the brain where myelin is destroyed.

- (a) The photograph shows lesions in the brain of a person with MS, as seen using magnetic resonance imaging (MRI).



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The actual length of this brain is 15 cm.

Calculate the actual diameter, in mm, of the lesion between X and Y.

(2)

Answer mm

- (b) The table shows the speed of conduction of an impulse along non-myelinated and myelinated neurones with axons of different diameters.

Type of neurone	Diameter of axon / μm	Speed of conduction of impulse / m s^{-1}
non-myelinated	1	2.3
non-myelinated	2	3.1
non-myelinated	4	4.2
myelinated	1	2.6
myelinated	2	8.5

- (i) Show that the effect of diameter on the speed of conduction of an impulse is six times greater in a myelinated neurone than in a non-myelinated neurone.

(2)

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(ii) Explain the effect of the myelin sheath and the diameter of the axon on the speed of conduction of an impulse.

(4)

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(iii) Symptoms of MS can include blindness in one eye and lack of coordination.
Suggest why a person with MS could have these symptoms.

(3)

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(Total for Question 5 = 11 marks)